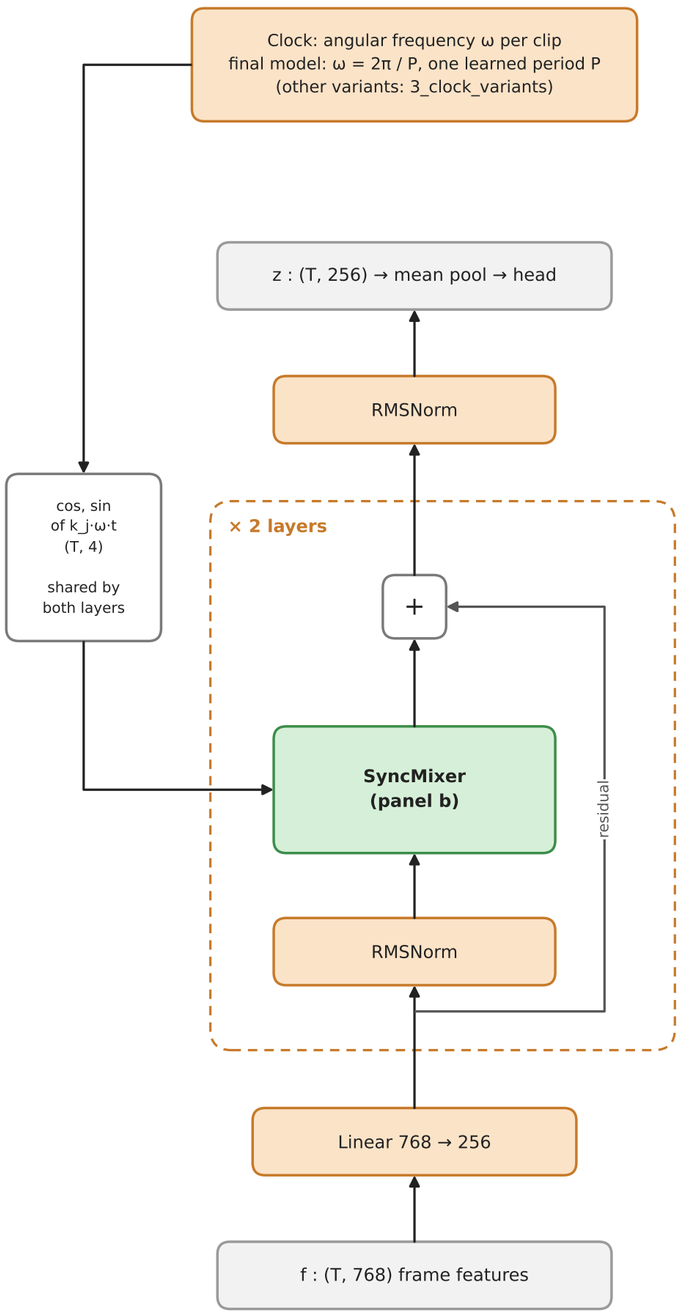
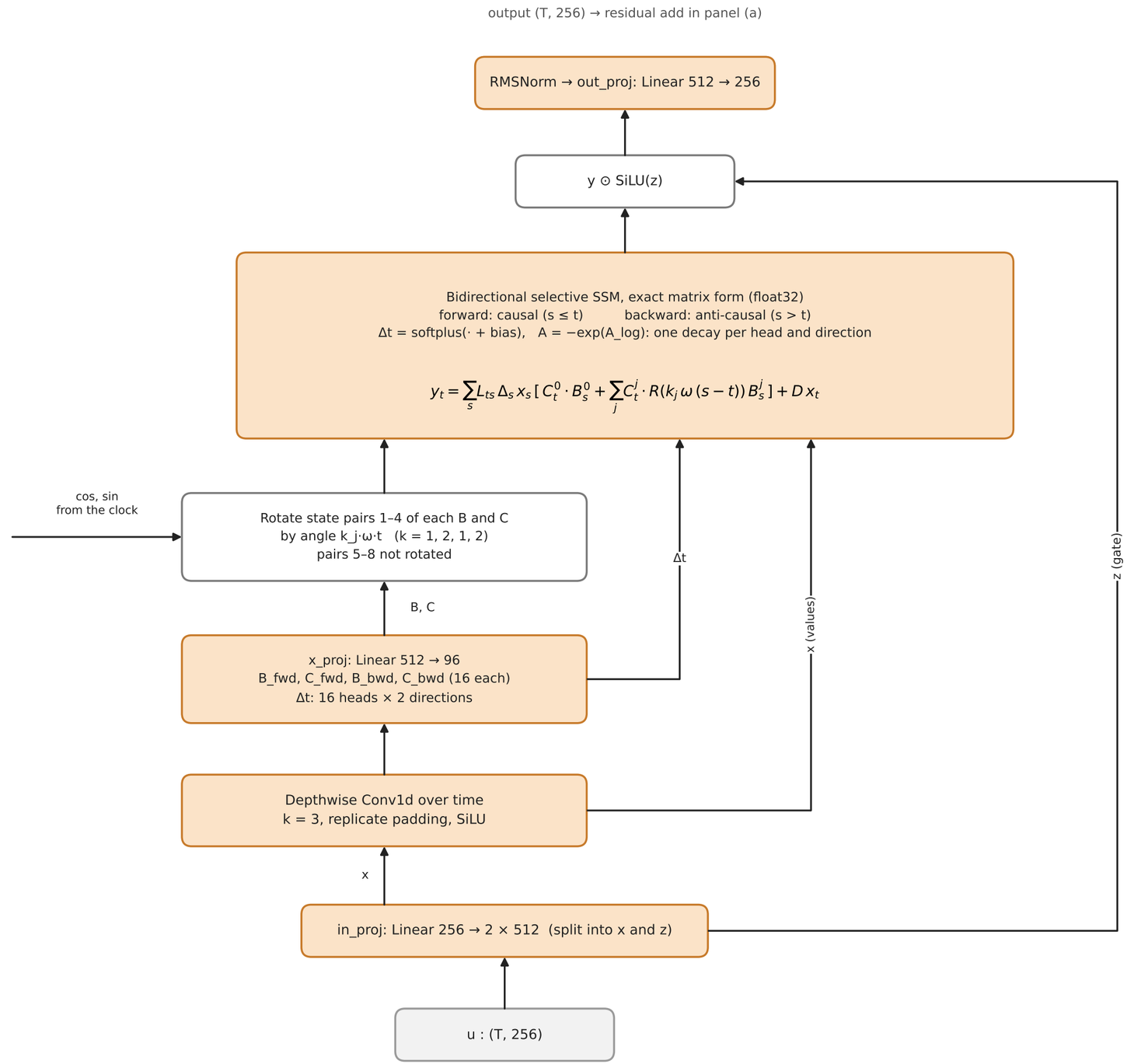
Sync temporal block (proposed)

trained · proposed block (trained) · no parameters · tensor

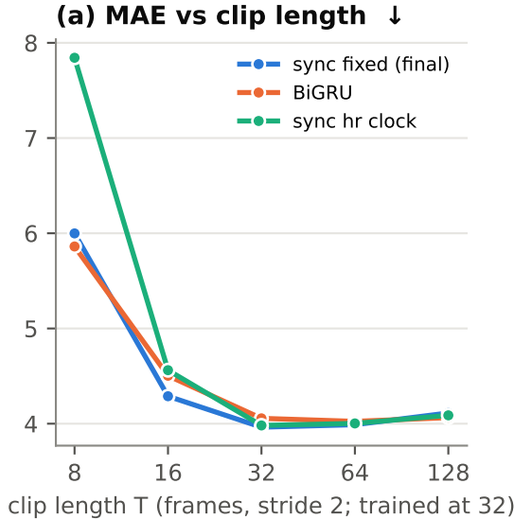
(a) SyncTemporal — 2 layers

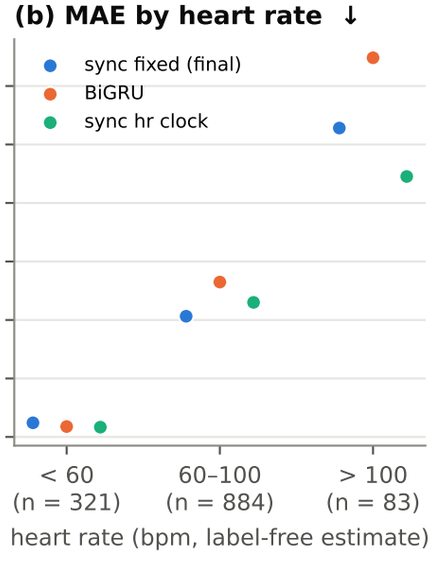


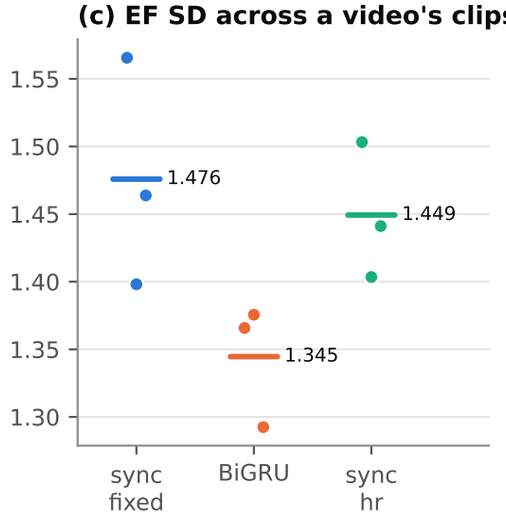
(b) SyncMixer — one layer: d_model 256, d_inner 512, 16 heads × 32 per direction, state N = 16

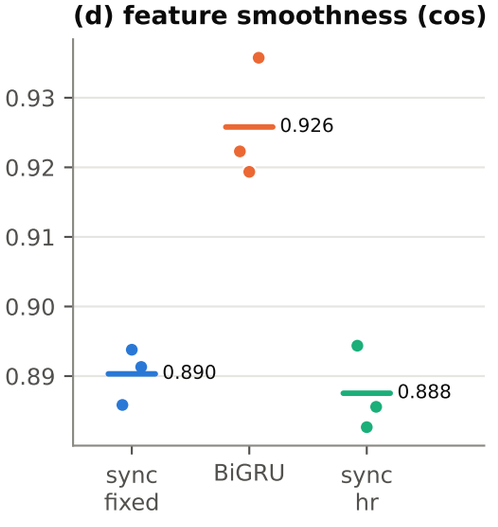


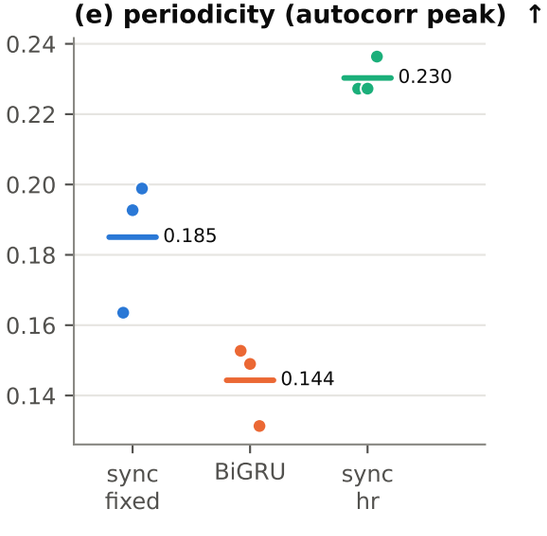
Sync block vs BiGRU on every evaluation metric — validation set (mean of 3 seeds; dots = seeds)

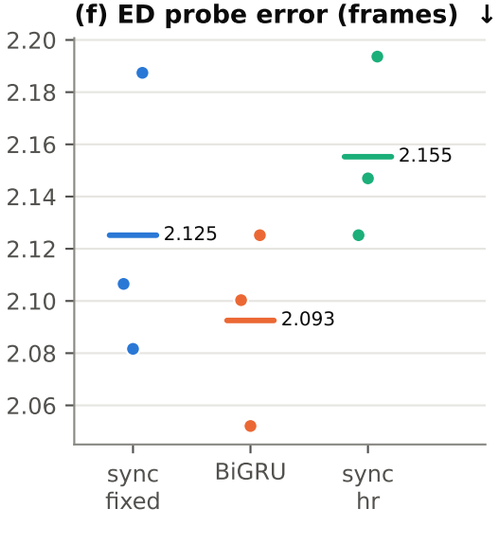


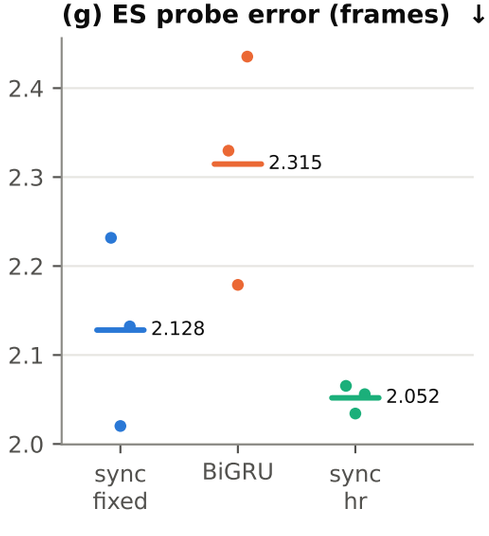










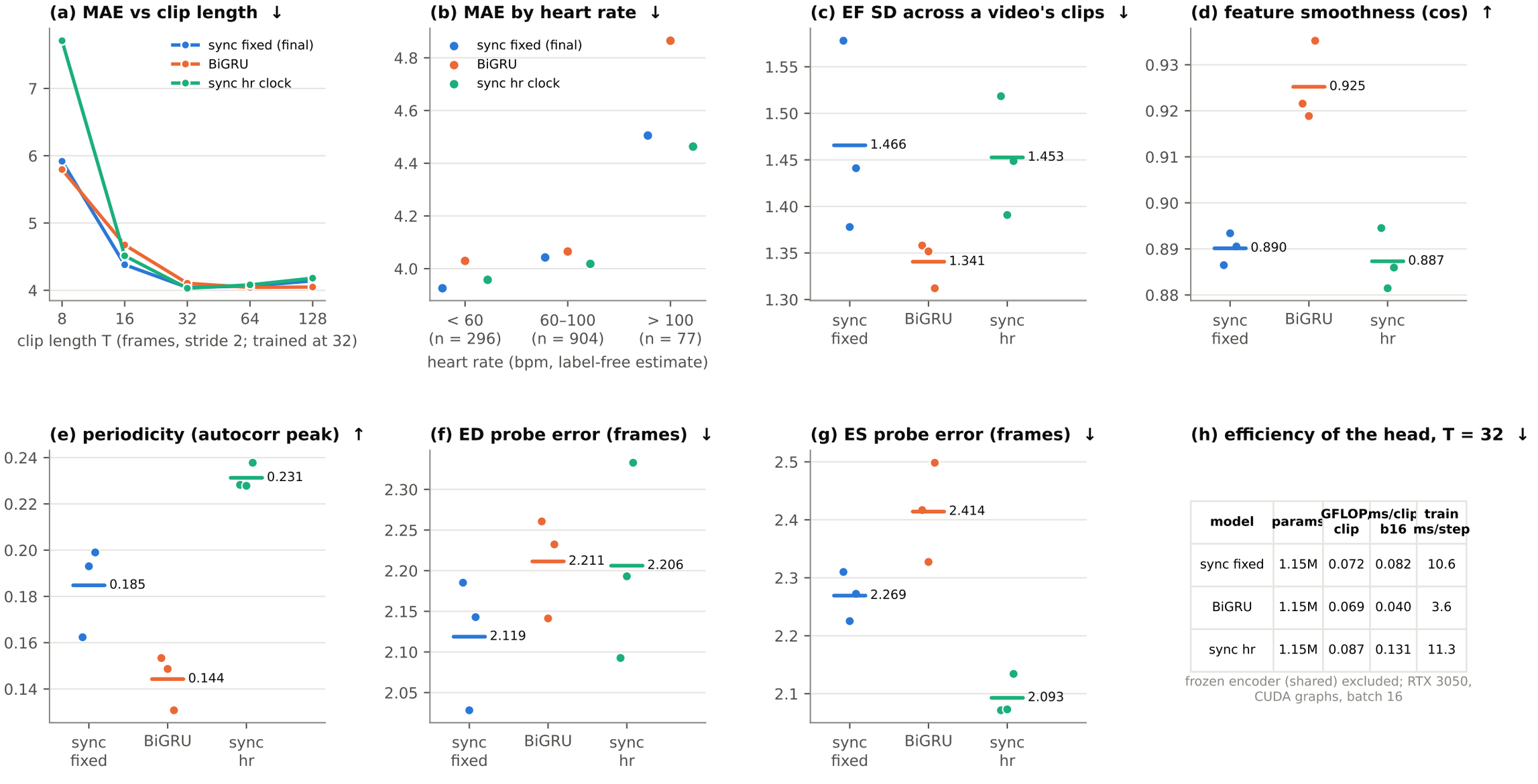


(h) efficiency of the head, T = 32 ↓

model	params	GFLOP/clip	ms/clip b16	train ms/step
sync fixed	1.15M	0.072	0.082	10.6
BiGRU	1.15M	0.069	0.040	3.6
sync hr	1.15M	0.087	0.131	11.3

frozen encoder (shared) excluded; RTX 3050, CUDA graphs, batch 16

Sync block vs BiGRU on every evaluation metric — test set (mean of 3 seeds; dots = seeds)



(h) efficiency of the head, T = 32 ↓

model	params	GFLOP/clip	ms/clip b16	train ms/step
sync fixed	1.15M	0.072	0.082	10.6
BiGRU	1.15M	0.069	0.040	3.6
sync hr	1.15M	0.087	0.131	11.3

frozen encoder (shared) excluded; RTX 3050, CUDA graphs, batch 16